

Albers Aerospace Comprehensive Company Overview

1. Introduction

Company Overview Albers Aerospace stands at the forefront of the aerospace and defense industry, offering a wide array of solutions that cater to the evolving needs of our clients. Our commitment to innovation, quality, and excellence drives us to deliver high-value services and products that significantly enhance operational capabilities across various sectors. Our expertise is deeply rooted in multiple domains, including defense, industrials, innovative technologies, and unmanned aerial systems.

Business Verticals:

- **Defense Solutions:** Albers Aerospace provides a comprehensive suite of defense solutions, ranging from tactical aircraft support and unmanned aviation operations to advanced sense and avoid technologies. The integration of IOMAX into our operations has bolstered our capabilities in ISR and light attack aircraft, allowing us to deliver specialized aircraft solutions that meet the rigorous demands of modern defense operations.
- **Industrials:** Our industrial division is dedicated to precision machining, sheet metal fabrication, and the production of specialty storage containers. Facilities like Heritage Aviation and Garrett Container Systems enable us to deliver high-quality manufacturing services that support both military and commercial clients, ensuring reliability and excellence in every product.
- **Innovative Technologies:** At the cutting edge of technological advancement, we develop and implement state-of-the-art solutions, including advanced sensors, AI-driven systems, and model-based systems engineering. The acquisition of Defense Research Associates (DRA) has further strengthened our R&D capabilities, allowing us to innovate in areas such as radar systems, unmanned aircraft, and AI applications.
- **Unmanned Aerial Systems (UAS):** The recent acquisition of Skyfront LLC has expanded our UAS capabilities, introducing long-range and endurance drones into our portfolio. Skyfront's flagship products, such as the Perimeter 8 hybrid multirotor and the Lawn Dart UAS, offer advanced GPS-free navigation and electronic warfare resilience. These additions enhance our ability to provide versatile and robust UAV solutions for both defense and commercial markets.

By integrating the innovative technologies and expertise of IOMAX, Skyfront, and other strategic acquisitions, Albers Aerospace continues to lead in delivering exceptional aerospace and defense solutions. Our mission is to ensure the success of our customers' missions across diverse operational environments, maintaining a steadfast commitment to excellence and innovation.

Mission and Vision

- **Mission:** To deliver exceptional aerospace and defense solutions that ensure the success of our customers' missions.
- **Vision:** To be a leader in aerospace and defense, driving innovation and operational excellence across the industry.

Core Values

- **Integrity:** Upholding the highest standards of ethics and transparency.
- **Innovation:** Continuously pushing the boundaries of technology and processes.
- **Excellence:** Committed to delivering the highest quality in everything we do.

- **Teamwork:** Fostering collaboration and respect among our team and partners.
- **Customer Focus:** Prioritizing the needs and success of our customers.

2. Corporate Information

History and Milestones

- **2014:** Formation of The Albers Group.
- **2015-2017:** Expansion into systems engineering subcontract efforts supporting the US Navy and large primes.
- **2018:** Acquisition of Ground Sensor Tech and Albers Plating.
- **2019:** Achieved AS9100D certification.
- **2020:** Acquired Unmanned Systems Inc. and Heritage Aviation Ltd.
- **2021:** Rebranded as Albers Aerospace.
- **2022:** Acquisition of Garrett Container Systems.
- **2023:** Continued expansion with the acquisition of Defense Research Associates (DRA).

Organizational Structure

- **CEO:** John Albers
- **CEO:** John Albers
- **COO:** Franklin Byrd
- **CFO:** Chris Hansen
- **CRO:** Rich Carter
- **CTO & EVP Innovation:** Mike Frizzell
- **EVP Albers Defense:** Danny Johnson
- **EVP Albers Industrials:** Bruce Patterson
- **Business Units:**
 - Defense
 - Industrials
 - Innovative Technology

Certifications and Accreditations (Updated with Skyfront Integration)

1. **AS9100D Certification:** Albers Aerospace maintains AS9100D certification, ensuring the highest standards in quality management systems for aerospace manufacturing and services.
2. **FAA 145 Repair Station Certification:** Our facilities are certified as FAA 145 Repair Stations, allowing us to perform maintenance and alterations on a wide range of aircraft components.
3. **NADCAP Certification (in process):** We are in the process of obtaining NADCAP certification, which will further validate our commitment to excellence in aerospace manufacturing processes.

ITAR Registered: Albers Aerospace keeps safeguarding our warfighters and national security front of mind. Understanding how to navigate U.S. the International Traffic in Arms Regulations (ITAR) and the Export Administration Regulations (EAR) that govern the export of sensitive defense and dual-use technologies is critical to mission success. Albers is export-ready with a strong compliance environment, procedural controls, process rigor, and skilled people to meet complex requirements. Our team instills trust with our customers by having the right global trade authorizations whatever the situation in advance of mission requirements, allowing technology transfer and exports at the speed of need for the customer.

Supporting U.S. national security objectives through compliance and meeting regulatory standards enables us to continue to earn our customers' trust. The defense industry places a premium on reliability and security, particularly when it comes to handling controlled technologies. Albers Aerospace has a strong track record of regulatory compliance across its enterprise and maintains robust internal controls, secure supply chains, and ethical business practices.

At Albers Aerospace, we believe compliance also fosters innovation and more efficient program management. By integrating export control considerations early in the research and development phase, Albers can streamline approval processes, minimize project delays, and avoid reengineering products to meet regulatory demands. This foresight enhances Albers' time-to-market capabilities, allowing more rapid response to emerging needs and opportunities in both domestic and international defense sectors.

Alignment with U.S. national security policy and defense priorities is in Albers Aerospace DNA. At Albers we understand that it is not just about technical performance, it's about being a trusted partner in safeguarding U.S. technology and interests.

- **Skyfront Certifications:** Skyfront, now part of Albers Aerospace, holds a CAGE Code of 8D702 and a UEI of WMDZZUA9J1R3. They are registered with the Small Business Innovation Research (SBIR) program, which supports their innovative UAV technologies for defense and commercial applications.

Company Details (Updated for IOMAX and Skyfront Acquisitions)

1. **Company:** Unmanned Systems, Inc. doing business as Albers Aerospace
2. **Nickname:** USI dba Albers Aerospace, Albers Aerospace, Albers Aero
3. **CAGE:** 3FGY3
4. **DUNS:** 131010865
5. **TIN:** 27-0056512
6. **UEI:** LQYFB4Y5LA57
7. **SBA SBC ID Number:** 2269675
8. **Address:** 1700 Redbud Blvd STE 130 #400, McKinney, TX 75069
9. **Size:** Small - most NAICS as of May 2023
10. **State of Org:** Nevada
11. **Entity Info:** Registered as foreign entity doing business in Texas
12. **Parent of Heritage, GCS, IOMAX, and Skyfront**
13. **ITAR Cert:** Yes, includes GCS, expires 07/2024

NAICS Codes:

- **Aerospace Manufacturing:**
 - 336412: Aircraft Engine and Engine Parts Manufacturing
 - 336413: Other Aircraft Parts and Auxiliary Equipment Manufacturing
 - 332710: Machine Shops (for precision machining of sheet metal, titanium, Inconel)
- **Aviation Maintenance, Repair, and Overhaul (MRO):**
 - 488190: Other Support Activities for Air Transportation (which includes aviation CLS)
 - 611512: Flight Training (for Group 5 Unmanned Systems Training, Operations, Maintenance)
- **Engineering Solutions:**
 - 541330: Engineering Services (for Model-Based Systems Engineering, Testing, DO-178C Compliance)
 - 541715: Research and Development in the Physical, Engineering, and Life Sciences (except Nanotechnology and Biotechnology)
- **ALBERS Defense:**
 - 334511: Search, Detection, Navigation, Guidance, Aeronautical, and Nautical System and Instrument Manufacturing (for intelligence surveillance sensors)
 - 334220: Radio and Television Broadcasting and Wireless Communications Equipment Manufacturing (for 4G LTE and edge computing)
- **Innovative Technologies:**
 - 541711: Research and Development in Biotechnology (for aerospace technology incubation)
 - 336411: Aircraft Manufacturing (for eVTOL design, development, test, and manufacturing)
- **Tactical Aircraft Solutions (Including IOMAX and Skyfront):**
 - IOMAX's integration into Albers Aerospace enhances capabilities in delivering specialized aircraft solutions, particularly in ISR and light attack aircraft, leveraging IOMAX's expertise in transforming commercial aircraft into ISR platforms and integrating advanced avionics and mission systems.
 - Skyfront's integration brings advanced UAV capabilities, including the Lawn Dart UAS, a long-range, heavy-lift loitering munition designed for GPS/RF-denied environments, enhancing our offerings in unmanned aerial systems.
 - 336411: Aircraft Manufacturing (for UAV design, development, test, and manufacturing)
- 541715: Research and Development in the Physical, Engineering, and Life Sciences (for UAV technology development and testing)

3. Capabilities by Vertical

3.1 Defense

Tactical Aircraft Solutions (Enhanced with IOMAX and Skyfront Capabilities)

1. **Capabilities:**
 - **Developmental Test and Operational Flight Support:** Albers Aerospace, now incorporating the expertise of IOMAX and Skyfront, offers comprehensive support for test and operational flight activities. This includes preparation, execution, and analysis, ensuring that aircraft and unmanned systems are mission-ready and compliant with all regulatory standards.
 - **Aircraft Maintenance and Modification:** With IOMAX's proven track record in aircraft modification and integration, and Skyfront's cutting-edge UAV technology, we provide unparalleled expertise in performing maintenance, modifications, and upgrades on various aircraft and unmanned platforms. This ensures optimal performance and compliance with regulatory standards, leveraging IOMAX's experience in transforming commercial aircraft into ISR platforms and Skyfront's innovations in GPS-free navigation and electronic warfare resilience.
 - **Logistics Support:** We offer end-to-end logistics services to ensure the availability and readiness of aircraft and related systems. This includes the integration of IOMAX's capabilities in delivering

comprehensive logistical support packages, including maintenance, training, and sustainment services, as well as Skyfront's advanced UAV solutions for long-range and endurance missions.

2. Key Projects:

- **Aircraft Intermediate Maintenance Department (AIMD) - U.S. Navy:** Albers Aerospace, as a subcontractor to Vertex Aerospace, provides I-Level Maintenance, repair, and logistics support services at Naval Air Stations Corpus Christi and Pensacola. This includes labor, materials, and tools necessary for maintaining aircraft components, avionics equipment, and machine shop operations. The contract, recompleted successfully in 2022, ensures continued support through September 2028. Responsibilities encompass a wide range of tasks, such as aircraft component maintenance, avionics equipment upkeep, and incorporating I-Level technical directives. We also handle machine shop operations, calibration, Non-Destructive Inspection, and more.
- **IOMAX ISR Aircraft Integration:** IOMAX has successfully integrated ISR capabilities into various aircraft, including the Cessna Caravan and Air Tractor platforms. This includes the integration of advanced avionics, EO/IR sensors, and weapons management systems, enhancing the operational capabilities of our clients. Recent projects include the transformation of Cessna Caravans for the Royal Jordanian Air Force, showcasing our ability to deliver complex engineering solutions on time and within budget.
- **Skyfront Lawn Dart UAS Development:** Skyfront's Lawn Dart UAS is a long-range, heavy-lift loitering munition designed to operate autonomously without GPS in RF-contested environments. This innovative UAV offers a 100-mile range while carrying a 45 lb. payload, providing the USAF and other defense clients with a cost-effective, electronic warfare-resilient solution for modern combat scenarios.

Unmanned Aviation Operations (Group 5 UAVs, MQ-9 Training)

1. Capabilities:

- **MQ-9 Aircrew Training and Operations:** Managed a fully Government-Owned, Contractor-Operated (GOCO) training facility at TSTC Waco for MQ-9 aircrew, providing both long-range and mission control element operations training. Our ISR training and flight training allowed for successful GOCO operations around the world in support of OSD priorities.
- **ISR Training and Flight Operations:** Conducted comprehensive ISR training, ensuring high standards of safety and operational effectiveness. Our ISR training and flight training ensured high-quality support and continual process improvement.
- **Launch and Recovery Element (LRE) Support:** Provided qualified MQ-9 pilot and sensor operators for LRE operations, supporting real-world missions and training exercises. Performance met contractual requirements and exceeded many to the Government's benefit.

2. Key Projects:

- **MQ-9 Training Facility at TSTC Waco:** Albers Aerospace managed this GOCO facility, achieving a 100% training completion rate for over 200 students. The facility supported various operations, including flight and simulator events, with efficient scheduling processes that minimized travel and labor costs for the government. During rapid increases in flight operations, Albers efficiently maintained staffing levels and minimized administrative overhead by utilizing instructor cadre for additional tasks, reducing overall costs for the government.

Maintenance, Repair, and Overhaul (MRO)

• Capabilities:

- **Comprehensive Aircraft Maintenance:** Full-spectrum maintenance services, including preventive and corrective maintenance, to ensure aircraft readiness and reliability.

- **Component Overhaul and Modification:** Expertise in overhauling and modifying aircraft components to extend their operational life and enhance performance.
- **Logistics and Supply Chain Management:** Integrated logistics support, including parts procurement, inventory management, and supply chain optimization.
- **Key Projects:**
 - **NASA Aircraft Operations Support:** As a subcontractor to Vertex Aerospace, Albers Aerospace provides comprehensive support for NASA's Aircraft Operations at the Armstrong Flight Research Center. This includes maintenance, modifications, and flight-line operations for a diverse fleet of aircraft, ensuring adherence to environmental and safety regulations. Our support extends to aircraft maintenance, modifications, and flight-line operations. We handle all aspects of operations for two Armstrong Chemical/Tool cribs, including requisitioning, inventory control, and adherence to environmental and safety regulations. Additionally, we oversee operations for two Experimental Fabrication shops, which involve hydraulic, machine, sheet metal, welding, and production control functions.

Defense Contracts and Services

1. **Key Partnerships:**
 - **Vertex Aerospace:** Albers Aerospace has maintained a long-term collaboration with Vertex Aerospace on various defense and aerospace projects. This includes significant roles in the Aircraft Intermediate Maintenance Department (AIMD) and providing comprehensive support for NASA Aircraft Operations at the Armstrong Flight Research Center. These projects highlight our capability to deliver high-quality maintenance, repair, and logistics support services.
 - **General Atomics:** We have partnered with General Atomics to provide MQ-9 pilot support for classified operations. This partnership underscores our expertise in unmanned aviation operations and our ability to support complex, sensitive missions.
 - **NASA:** Albers Aerospace has been instrumental in providing aircraft operations support at NASA's Armstrong Flight Research Center. Our work ensures that NASA's diverse fleet of aircraft is maintained to the highest standards, supporting their critical research and operational missions.
 - **USAF:** We have delivered advanced sense and avoid technologies for unmanned aircraft systems to the United States Air Force. This contribution enhances the safety and operational effectiveness of USAF's unmanned systems, showcasing our commitment to innovation and excellence in aerospace technology.
 - **Northrop Grumman:** Our collaboration with Northrop Grumman has focused on implementing lean process improvement initiatives. These efforts have resulted in significant labor and cost savings, demonstrating our ability to enhance efficiency and productivity in aerospace manufacturing and operations.
 - **IOMAX:** With the recent acquisition of IOMAX, Albers Aerospace has expanded its capabilities in delivering specialized aircraft solutions, particularly in ISR and light attack aircraft. IOMAX's expertise in transforming commercial aircraft into ISR platforms and integrating advanced avionics and mission systems further strengthens our position in the defense sector.

3.2 Industrials

Garrett Container Systems

- **Capabilities:**

- **Engineering and Manufacturing:** Specializing in the production of aluminum weapons and specialty storage containers, offering precision machining and sheet metal fabrication.
- **Assembly and Manufacturing Services:** AS9100D registered services, including component and wiring harness assembly, welding, painting, and chromate conversion coating.
- **Lean Process Improvement:** Implementing lean methodologies to enhance manufacturing efficiency and reduce costs.
- **Key Projects:**
 - **Weapons and Specialty Storage Containers:** Provided engineering and manufacturing services for a variety of military and commercial clients, delivering high-quality, precision-engineered products. Garrett Container Systems specializes in the design, manufacturing, and supply of aluminum weapons and specialty storage containers for various military and commercial clients. They offer precision machining, sheet metal fabrication, and welding services. Additionally, they provide lean process improvement to enhance manufacturing efficiency and reduce costs.

Aerospace Manufacturing

- **Capabilities:**
 - **Precision Machining:** Utilizing advanced CNC machining techniques to produce high-precision components for aerospace applications. Heritage Aviation specializes in metal fabrication where cutting-edge custom CNC machining is one of their strengths. They have developed the machining expertise and experience to handle all CNC machining needs. Utilizing Mastercam, SolidWorks, and Esprit maximum capability, they offer precision machining for metals and plastics, component and wiring harness assembly, and sheet metal fabrication.
 - **Sheet Metal Fabrication:** Offering comprehensive sheet metal fabrication services, including cutting, forming, and assembly. Heritage Aviation's sheet metal fabrication department combines over 50 years of expertise with high-tech machinery and software to take on the most complex parts. They offer services including precision manufacture of components, sheet metal fabrication, and custom CNC machining.
 - **Component and Wiring Harness Assembly:** Expertise in assembling complex wiring harnesses and components for aerospace systems. Heritage Aviation's component and wiring harness assembly capabilities include highly trained and experienced workers capable of building any type of wire harness bundle, braiding machines for expanded capabilities, and IPC J-STD-001 certification.
 - **Special Processing Capabilities:** Providing welding, painting, and chromate conversion coating services to meet stringent aerospace standards. Heritage Aviation offers surface finishing services, including paint booth applications, MIL-C-8514, MIL-C-85285, MIL-PRF-23377, MIL-PRF-85285 accreditations, and an aluminum treatment system for chromate conversion coating.
- **Key Equipment:**
 - **DMG DMU75 Simultaneous 5 Axis:** Capable of machining large parts with high precision, designed for various metals.
 - **DMG DuraTurn 2050 Lathe:** High-performance lathe with diverse machining options.
 - **Agie Charmiles Cut 20P Wire EDM:** Advanced wire EDM for precise cutting of complex shapes.

- **DMG DMU50 3+2 Positional 5 Axis:** Ideal for complex parts, designed for light steel and aluminum.
- **Fadal VCM 2216 3 Axis:** Capable of handling small to medium part manufacturing.
- **Trumpf TruMatic 2020R Punch:** Allows for fast and precise punching, holding tight tolerances.
- **YCM 108B:** Designed for hard material cutting with 3-axis machining and 4th-axis addition.
- **Baileigh Roll Former:** Hydraulic plate roll former designed for 60" bending length.

Lean Process Improvement

- **Capabilities:**

- **Lean Services:** Expertise in conducting lean events and training sessions to identify and eliminate waste, improve processes, and enhance efficiency.
- **Collaborative Assessments:** Working with clients to assess current processes and implement improvements that increase productivity and reduce costs.

- **Key Projects:**

- **Northrop Grumman:** Conducted lean process improvement initiatives that resulted in labor savings of over 5000 hours and \$1.6M per Air Vehicle (AV). Conducted numerous events and training sessions resulting in labor savings of over 5000 hours and \$1.6M per Air Vehicle (AV).
- **Vertex Aerospace:** Improved T-45 maintenance lines at Naval Air Station Kingsville, increasing the number of ready aircraft from 41 to 54 in two months. Provided a collaborative assessment of T-45 maintenance lines with the goal of increasing the number of ready aircraft on the flight line. The engagement resulted in an increase of ready aircraft from 41 to 54 in a two-month period.
- **Garrett Container Systems:** Achieved a 10% improvement in manufacturing efficiency and prepared for critical certifications. Current efforts focus on preparing GCS for a critical certification, resulting in a 10% improvement in efficiency on the manufacturing line over three months. Additionally, the team is in discussion with Boston Consulting Group to provide CPI to the US Navy's Aviation Training Command, which could result in a multi-year effort.

3.3 Innovative Technology

Defense Research Associates (DRA)

- **Capabilities:**

- **Research and Development:** Expertise in developing advanced technologies for unmanned aircraft, radar systems, and AI. DRA offers research, development, fabrication, rapid prototyping, and product design capabilities over a range of technical specialties. DRA's capabilities include advanced technologies, unmanned aircraft, counter unmanned aircraft, radar warning systems, laser warning systems, artificial intelligence, and electro-optical/infrared sensors.
- **Rapid Prototyping and Product Design:** Capabilities to quickly develop and test prototypes, ensuring rapid innovation and deployment. DRA performs hardware and software engineering, mechanical design and fabrication, systems characterization and testing, reverse-engineering, and systems exploitation in both unclassified and classified environments.

- **Systems Engineering:** Providing comprehensive systems engineering services, including hardware and software development, mechanical design, and testing. DRA has over 30 years of experience delivering DoD capabilities and offers organic laboratories and facilities, including systems integration, optics, test and integration, and system processing and analysis labs.
- **Key Projects:**
 - **Air Force Unmanned Aircraft Sense and Avoid:** Developed and implemented advanced sense and avoid technologies for unmanned aircraft systems, significantly enhancing operational safety and effectiveness. This project involved rigorous testing and integration of the technology into existing Air Force systems, ultimately contributing to a notable improvement in safety and operational efficiency. By delivering sophisticated detection and avoidance systems, DRA ensured compliance with stringent Air Force standards and operational requirements.

Model-Based Systems Engineering (MBSE)

- **Capabilities:**
 - **Ground Control Station Software for UAS:** Development of advanced software solutions to enhance the control and efficiency of unmanned systems. Developed ground control station software for UAS, enhancing control and efficiency.
 - **Cockpit Technology Refresh:** Modernizing cockpit technology to improve pilot interfaces and system integration. Implemented cockpit technology refresh programs to improve pilot interfaces and system integration.
 - **Digital Backbone:** Developing robust digital infrastructures for rotary-wing platforms and other systems. Developed digital backbones for rotary-wing platforms to ensure seamless integration and system communication.
 - **Health Awareness and Monitoring:** Implementing systems to monitor and report on the health and performance of various aerospace systems. Developed health awareness and monitoring systems for various aerospace systems, ensuring real-time monitoring and reporting.
- **Key Projects:**
 - **Ground Control Station Software for UAS:** Developed and deployed advanced software solutions that significantly improved the control and efficiency of unmanned systems, enhancing operational capabilities and situational awareness. Developed ground control station software for UAS, enhancing control and efficiency. Provided robust digital infrastructures for rotary-wing platforms and other systems.

Sensors and AI Technologies

- **Capabilities:**
 - **Missile Detection and Tracking:** Development of advanced sensors and algorithms for missile detection and tracking, ensuring rapid and accurate threat identification. Developed missile detection and tracking systems for enhanced threat identification.
 - **Machine Learning Algorithms:** Leveraging AI and machine learning to enhance target detection, classification, and situational awareness. Developed machine learning algorithms for target detection and classification.

- **Imaging Systems:** Developing cutting-edge imaging systems to improve operational effectiveness and safety. Developed advanced imaging systems for improved operational effectiveness.
- **Key Projects:**
 - **DARPA Small UAS Cued Imaging & Machine Learning Classification:** Created machine learning algorithms and imaging systems for small unmanned aircraft systems, focusing on target detection and classification. The systems improved situational awareness and operational effectiveness, showcasing the ability to innovate and deliver complex technological solutions. Leveraged AI to enhance target detection and classification.

Emerging Technologies and Innovations

- **Capabilities:**
 - **Hybrid VTOL (Group 2 & 3):** Development and deployment of hybrid vertical takeoff and landing systems to enhance operational flexibility and efficiency. Developed hybrid VTOL systems for improved operational flexibility.
 - **Ultra-Short Pulse Laser (USPL) Paint Ablation & Removal:** Utilizing USPL technology for precise and efficient paint removal and surface preparation. Developed USPL technology for paint ablation and removal.
 - **Composite Laser Ablation for Surface Preparation (CLASP):** Advanced laser technology for surface preparation and treatment. Developed CLASP technology for surface preparation and treatment.
- **Key Projects:**
 - **Hybrid VTOL Development:** Advanced the development of hybrid VTOL technologies, resulting in significant improvements in operational capabilities and flexibility. Developed hybrid VTOL technologies to enhance operational capabilities.

4. Past Performance (Updated with IOMAX Integration)

Key Projects and Contracts

Aircraft Intermediate Maintenance Department (AIMD) - U.S. Navy

1. **Description:** Providing I-Level Maintenance, repair, and logistics support for Naval Air Stations Corpus Christi and Pensacola. Responsibilities include aircraft component maintenance, avionics equipment upkeep, machine shop operations, calibration, Non-Destructive Inspection, and more.
2. **Outcome:** Enhanced readiness and mission capability, ensuring the operational effectiveness of the supported activities. Successfully recompeted the contract in 2022, securing continued support through September 2028.

Aircraft Operations Support - NASA

1. **Description:** Comprehensive support for various NASA aircraft at the Armstrong Flight Research Center, including maintenance, modifications, and flight-line operations. Responsibilities include managing the operations of aircraft ranging from motor gliders to large commercial, military, and experimental research aircraft.

2. **Outcome:** Ensured operational efficiency and compliance with environmental and safety regulations, contributing to the success of NASA's research and support missions.

Maintenance Modification, Aircrew and Related Services - Navy Test Wing Atlantic

1. **Description:** Providing full-spectrum maintenance support for NAWC T&E assets, including C-130 aircraft. Services encompass Organizational Level (O-Level), Intermediate Level (I-Level), and Depot Level (D-Level) maintenance, ensuring the assets meet the required standards for test schedules.
2. **Outcome:** Met rigorous standards required for flight and ground test schedules, enhancing the operational readiness and effectiveness of the Navy Test Wing Atlantic.

MQ-9 Flight Operations - Classified Customer

1. **Description:** Providing MQ-9 pilots for flight operations supporting a highly classified customer. Responsibilities included training, operational support, and ensuring compliance with all applicable regulations.
2. **Outcome:** Supported real-world operations with a 100% completion rate for training, demonstrating the ability to adapt to dynamic requirements and ensure mission success.

Small & Medium Remotely Piloted Aircraft (RPA) Pilot Services

1. **Description:** Providing RPA pilot services for various commercial and government agencies. Services include integration of unmanned air vehicles, flight test data collection, flight characteristics validation, and modifications to existing UAVs or new aircraft designs.
2. **Outcome:** Consistently delivered high-quality services, resulting in continuous contracts and client satisfaction. Supported multiple agencies, including the FAA, Department of Energy, and various commercial clients.

IOMAX Key Projects

Cessna Caravan ISR Integration

1. **Description:** Integration of ISR mission payloads for the Royal Jordanian Air Force, including advanced avionics and communication systems.
2. **Outcome:** Successfully transformed Cessna Caravans into ISR platforms, enhancing operational capabilities with advanced systems and training.

Acquisition of Air Tractor AT-802 Firefighting Aircraft

1. **Description:** Procurement and delivery of firefighting aircraft for the Royal Jordanian Air Force, meeting urgent regional firefighting needs.
2. **Outcome:** Delivered aircraft on schedule, providing critical aerial firefighting capabilities and training services.

Upgrade of AT-802 Border Patrol Aircraft

1. **Description:** Upgrading six AT-802i aircraft for the Royal Jordanian Air Force with advanced systems and capabilities.
2. **Outcome:** Enhanced surveillance capabilities and extended the operational life of the aircraft, ensuring continued mission success.

Detailed Project Descriptions

Air Force Unmanned Aircraft Sense and Avoid

1. **Objectives:** Enhance operational safety and effectiveness of unmanned aircraft systems.
2. **Scope of Work:** Develop and implement advanced sense and avoid technologies, conduct rigorous testing, and integrate the technology into existing Air Force systems.
3. **Outcomes:** Improved safety and operational efficiency, ensuring compliance with stringent Air Force standards.

Air Force Software Defined Radios

1. **Objectives:** Enhance communication capabilities across various operational environments.
2. **Scope of Work:** Engineer and deploy software-defined radios, providing state-of-the-art radio systems with enhanced flexibility and interoperability.
3. **Outcomes:** Delivered high-performance radio systems that met technical and operational standards, completed on time and within budget.

Navy Missile Warning System

1. **Objectives:** Enhance threat detection and response capabilities.
2. **Scope of Work:** Design and test advanced missile warning systems, deliver cutting-edge sensors and software solutions.
3. **Outcomes:** Successfully integrated systems into naval platforms, significantly improving mission readiness and safety.

DARPA Small UAS Cued Imaging & Machine Learning Classification

1. **Objectives:** Enhance target detection and classification for small unmanned aircraft systems.
2. **Scope of Work:** Develop machine learning algorithms and imaging systems, conduct testing and integration.
3. **Outcomes:** Improved situational awareness and operational effectiveness, showcasing the ability to innovate and deliver complex technological solutions.

USN/USMC Support Services

1. **Objectives:** Ensure operational efficiency and readiness for various defense programs.
2. **Scope of Work:** Provide extensive engineering and logistics support services, including support for the Common Aviation Command & Control System (CAC2S) and Tactical Air Control Party (TACP).
3. **Outcomes:** Maintained high levels of service quality and operational support, contributing to the success of multiple critical defense programs.

ESTEEM-TO 1

1. **Objectives:** Expedite the analysis of radar test events.
2. **Scope of Work:** Develop a Natural Language Processing (NLP) model for RF Spectrum Analysis, transforming complex radar data into natural language descriptions.
3. **Outcomes:** Established a proof of concept that laid the groundwork for future AI applications in radar signal analysis.

Threat Warning Technology Commercialization

1. **Objectives:** Develop and commercialize a missile warning system using Fourier Imaging System Technology.
2. **Scope of Work:** Create advanced software and firmware solutions, conduct comprehensive sensor testing and analysis.
3. **Outcomes:** Successfully tested and deployed the system, enhancing threat detection capabilities for the Air Force.

Laser Identification System (LasIdS)

1. **Objectives:** Detect and characterize laser energy directed at pilots to enhance safety.
2. **Scope of Work:** Develop and implement Laser Identification Systems, providing real-time alerts and data on laser threats.
3. **Outcomes:** Improved pilot safety, with the system successfully implemented and tested.

STAR Dynamics Digital Engineering

1. **Objectives:** Support radar product development for STAR Dynamics.
 2. **Scope of Work:** Provide comprehensive hardware and software engineering services, supporting the development and testing of advanced radar systems.
- **Outcomes:** Successfully completed key milestones, contributing to the advancement of STAR Dynamics' radar technologies.

5. Strategic Partnerships and Alliances

Key Partners and Collaborations

- **Vertex Aerospace:** Long-term collaboration on various defense and aerospace projects.
- **General Atomics:** Partnered on MQ-9 pilot support for classified operations.
- **NASA:** Provided aircraft operations support at the Armstrong Flight Research Center.
- **USAF:** Delivered advanced sense and avoid technologies for unmanned aircraft systems.
- **Northrop Grumman:** Implemented lean process improvement initiatives, resulting in significant labor and cost savings.

Joint Ventures and Teaming Agreements

- **Amentum:** Teamed on RPA O&M contract, providing qualified pilots and sensor operators.
- **Boston Consulting Group (BCG):** Collaborating on lean services for the US Navy's Aviation Training Command.

6. Contact Information (Updated with IOMAX Integration)

Corporate Contacts

1. **CEO:** John Albers
 - **Contact Information:** [Contact details not provided]
2. **COO:** Franklin Byrd

- **Contact Information:** [Contact details not provided]
- 3. **Business Development:** [Name, contact information not provided]

DCAA, ACO, and DFAS Office Contacts

1. **DCAA Office:** North Texas Branch Office
 - **POC:** Latrenda Garner
 - **Email:** latrenda.s.garner.civ@mail.mil
2. **ACO/DCMA Office:** DCMA Palmdale
 - **POC:** Teodorico Presbitero
 - **Email:** teodorico.c.presbitero.civ@mail.mil
3. **DFAS Office:** Columbus Center – Western Entitlement Operations
 - **POC:** N/A

IOMAX Contacts

1. **Chief Operating Officer:** Seamus Flatley
 - **Office:** 133 River Park Road, Mooresville, NC 28117
 - **Email:** sflatley@iomax.net
 - **Office Phone:** (704) 662-1891
 - **Mobile:** +1 (757) 769-6789, +1 (980) 435-7244

Appendices

Detailed Equipment Lists

- **Aerospace Precision Machining Equipment:**
 - **DMG DMU75 Simultaneous 5 Axis:** Capable of machining large parts with high precision, designed for various metals.
 - **DMG DuraTurn 2050 Lathe:** High-performance lathe with diverse machining options.
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Facility Descriptions (Updated with IOMAX and Skyfront Integration)

1. **Heritage Aviation, Ltd.:**
 - **Location:** Grand Prairie, TX
 - **Capabilities:** Specializes in precision machining, sheet metal fabrication, and operates as an FAA 145 Repair Station. The facility is equipped to handle complex aerospace manufacturing tasks, ensuring high-quality production standards.
 - **Facility Size:** 40,000+ square feet.

2. **Garrett Container Systems:**

- **Location:** Accident, Maryland
- **Capabilities:** Focuses on the engineering and manufacturing of specialty storage containers. The facility is designed to support the production of high-quality, durable containers for various applications, including military and commercial use.
- **Facility Size:** [Size not provided]

3. **IOMAX USA, LLC:**

- **Location:** Mooresville, NC
- **Capabilities:** IOMAX is renowned for its expertise in the design, integration, and delivery of specialized aircraft solutions. The facility supports complex engineering design, modification, integration, and flight testing of ISR aircraft. IOMAX's capabilities include transforming commercial aircraft into ISR platforms, integrating advanced avionics, EO/IR sensors, and weapons management systems.
- **Facility Size:** [Size not provided]

4. **Skyfront LLC:**

- **Location:** Redwood City, CA
- **Capabilities:** Skyfront specializes in long-range and endurance drones for commercial and defense applications. Their facility supports the development and production of hybrid-electric UAVs, including the Perimeter 8 and Lawn Dart UAS, which are designed for GPS-free navigation and electronic warfare resilience.
- **Facility Size:** [Size not provided]

Albers Aerospace – Current AI Landscape (May 2025)

Albers Aerospace has taken a structured, secure-first approach to integrating artificial intelligence into business operations. AI capabilities are actively deployed across two key business units: the Business Operations Unit (BOU) and the Business Intelligence Unit (BIU). These tools are enhancing speed, consistency, and decision-making without compromising compliance boundaries.

Platform Overview

Three primary AI tools are in use:

- **GovDash:** A FedRAMP-compliant, enterprise-grade AI platform, GovDash serves as the core system for proposal generation, opportunity analysis, and business intelligence reporting. It integrates directly with Microsoft Word for structured draft creation, enables automated document parsing, and supports secure handling of CUI (though this capability is not currently activated by BOU/BIU).
- **ClickUp:** Used as the centralized workflow platform for non-sensitive business operations. ClickUp manages project tasks, tracks proposal assignments, and provides real-time dashboards for team performance. It is fully integrated into daily operations and serves as the primary UI for BD task coordination.
- **ChatGPT:** Employed exclusively for non-sensitive brainstorming, drafting, and prompt engineering. It assists with the creation of instructions and task logic that are later used in secure AI platforms. This tool accelerates general content creation but is never used with CUI or proprietary information.

Use Cases by Unit

Business Operations Unit (BOU):

- Automates proposal outlines and drafts
- Tracks opportunity intake and prioritization
- Coordinates assignments, deadlines, and project visibility
- Streamlines content generation for internal use (e.g., summaries, comms, SOP drafts)

Business Intelligence Unit (BIU):

- Manages request intake through a standardized submission form
- Automatically assigns tasks in ClickUp based on capacity
- Conducts research, analysis, and report generation entirely in GovDash
- Delivers outputs in Albers-branded templates for leadership briefings
- Typical deliverables include opportunity intel briefs, ROM pricing, competitor due diligence, and contract strategy reports

Current Impact

AI integration has led to:

- Proposal drafting time reductions of 50% or more
- BI report generation cycles reduced by 60–80%
- Greater visibility into workload, pipeline health, and capture metrics
- Improved content quality and process repeatability
- Clear compliance segmentation between secure and non-secure environments

In Development: AI Agent Use Case

A new automation initiative is underway to create an **Intake → Qualification AI Agent**. This agent will:

- Automatically ingest and interpret new RFPs
- Apply company-specific qualification criteria (NAICS codes, security clearance, set-aside eligibility)
- Flag risk factors (e.g., incumbents, compliance constraints)
- Generate a Go/No-Go summary report for decision makers
- Push results directly to ClickUp for action

This use case represents the next evolution in Albers' AI adoption: shifting from tool-based efficiency to agent-driven decision support.